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DATABASE SCHEMA DESIGN
CAREERTRACK: FULL-STACK JOB PORTAL & APPLICANT TRACKING SYSTEM
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Program: Full Stack Web Development / MERN Stack Training Program
Database Engine: MongoDB Atlas (NoSQL Document Store)
ODM Library: Mongoose
Database Name: job_portal

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1. ENTITY RELATIONSHIP OVERVIEW

The CareerTrack database consists of three core collections:
- Admins (Independent collection managing authenticated Employer accounts)
- Jobs (Independent collection storing active and closed job postings)
- Applications (Child collection linked to Jobs via a Many-to-One ObjectId reference on the `jobId` field)

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2. COLLECTION 1: ADMINS (Model: backend/models/Admin.js)

Stores registered employer and administrator login credentials.

Field Name	Data Type	Constraints / Validation	Description
_id	ObjectId	Primary Key, Auto-generated	Unique Admin ID
email	String	Required, Unique, Lowercase	Employer login email
password	String	Required (Hashed via Bcryptjs)	Encrypted password hash
role	String	Default: "employer"	User access role
createdAt	Date	Auto-generated timestamp	Account creation date
updatedAt	Date	Auto-generated timestamp	Last update timestamp

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3. COLLECTION 2: JOBS (Model: backend/models/Job.js)

Stores all job listings created by employers and displayed on the job board.

Field Name	Data Type	Constraints / Validation	Description
_id	ObjectId	Primary Key, Auto-generated	Unique Job Posting ID

title | String | Required, Trimmed | Job role title
company | String | Required, Trimmed | Hiring company name
location | String | Required, Trimmed | City or Remote status
salary | String | Optional / Default: "Competitive" | Compensation range
type | String | Enum: Full-time, Part-time, | Employment classification

| | Internship, Contract |
description | String | Required | Full role responsibilities
requirements | String | Required | Required skills & exp
status | String | Enum: Active, Closed | Controls public visibility

| | Default: "Active" |
createdAt | Date | Auto-generated timestamp | Date job was published
updatedAt | Date | Auto-generated timestamp | Last modified date

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4. COLLECTION 3: APPLICATIONS (Model: backend/models/Application.js)

Stores candidate submissions, resume paths, and hiring pipeline states.

Field Name	Data Type	Constraints / Validation	Description
_id	ObjectId	Primary Key, Auto-generated	Unique Application ID
jobId	ObjectId	Required, Ref: 'Job'	Foreign key to Jobs
name	String	Required, Trimmed	Candidate full name
email	String	Required, Lowercase	Candidate contact email
phone	String	Required	Candidate phone number
qualifications	String	Required	Academic background
experience	String	Required	Work experience summary
resume	String	Required (File path / URL)	Stored PDF/DOCX path
status	String	Enum: Submitted, Under Review,	Current ATS stage

| | Shortlisted, Selected, Rejected |
| | Default: "Submitted" |
notes | String | Optional, Default: "" | Internal recruiter notes
createdAt | Date | Auto-generated timestamp | Application timestamp
updatedAt | Date | Auto-generated timestamp | Last status update time

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5. INDEXING & DATA INTEGRITY RULES

- Unique Account Constraint: The `email` field in the Admins collection is
  indexed as unique to prevent duplicate employer registrations.
- Duplicate Application Prevention: A compound validation check on (`jobId` +

`email`) in ApplicationController.js blocks candidates from submitting
multiple applications to the same job posting.
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